深圳市鸿讯电子有限公司

Shenzhen Hongxun Electronics Co., Ltd

PRODUCT SPECIFICATION

Customer 客户名称	
Part NO. 产品型号	CL28CK230-18A
Product type 产品内容	Mode:Transmissive type :Normallywhite. TFT LCD Module LCD Module: Graphic 240RGB*320Dot-matrix
Remarks 备注栏	□APPROVAL FOR SEPCIFICATIONS ONLY ■APPROVAL FOR SEPCIFICATIONS AND SAMPLE
Signature by Customer: 客户确认签章	

制定	审核	核准

Contents

1.0 GENERAL DESCRIPTION

1.1 Introduction

CL28CK230-18A is a color active matrix TFT LCD module using amorphous silicon TFT's (Thin Film Transistors) as an active switching devices. This module has a 2.8 inch diagonally measured active area with QVGA resolutions (**240** horizontal by **320** vertical pixel array). Each pixel is divided into RED, GREEN, BLUE dots which are arranged in vertical stripe and this module can display 262K colors. The TFT-LCD panel used for this module is adapted for a low reflection and higher color type.

1.2 General Specification

Parameter	Specification	Unit	Remarks
Active area 可视区域	43.20(H) ×57.60(V)	mm	
Number of pixels 点阵数量	240(H) ×320(V)	pixels	
Pixel pitch 像素尺寸	0.18(H) × 0.18(V)	mm	
Outline Dimension 外形尺寸	50.00(H) × 69.20(V) × 3.50 (D)		With RTP
Pixel arrangement 像素排列	Pixels RGB stripe arrangement		
Display colors 颜色数量	262K		
Display mode 显示模式	Normally White		
Module Driving IC 模组驱动芯片	ST7789V		
Interface 接口方式	4-Line Serial interface		
Supply voltage 电压	+ 2.8/+3.3	V	
View Direction 视角	12 O’clock		

2.0 ELECTRICAL SPECIFICATIONS

2.1 Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit	Remarks
Power Supply Voltage	V_{DD}	-0.3	4.2	V	
Power Supply For LED	V_{LED}	-0.3	3.5	V	
Operating Temperature	T_{OP}	-20	+70	℃	
Storage Temperature	T_{ST}	-30	+80	℃	

2.2 Electrical Specifications

Parameter	Symbol	Values			Unit	Notes
		Min	Typ	Max		
Power Supply Input Voltage	VDD	2.5	2.8	3.3	V	
Logic Power Supply Input Voltage	IOVCC	1.65	1.8	3.3	V	
Power Supply Current	IDD	-	-	-	mA	
TFT Gate On Voltage	VGH	13	-	17	V	
TFT Gate Off Voltage	VGL	-12	-	-7	V	
TFT Common Electrode Voltage	VCOM	-	-	-	V	Note 1

Notes :

1. VCOM must be adjusted to optimize display quality, as Crosstalk and Contrast Ratio etc.

2.3 Backlight Characteristic

Item	Symbol	Min	Typical	Max	Unit
LED module Forward voltage	V_{LED}	2.9	3.1	3.3	V
LED module current	V_{LED}		80	100	mA
LCM Surface Luminance ★1	L_S	350			Cd/m³
LCM Surface brightness uniform★2	L_D	80			%

★1Test condition is:

(1) Center point on active area.

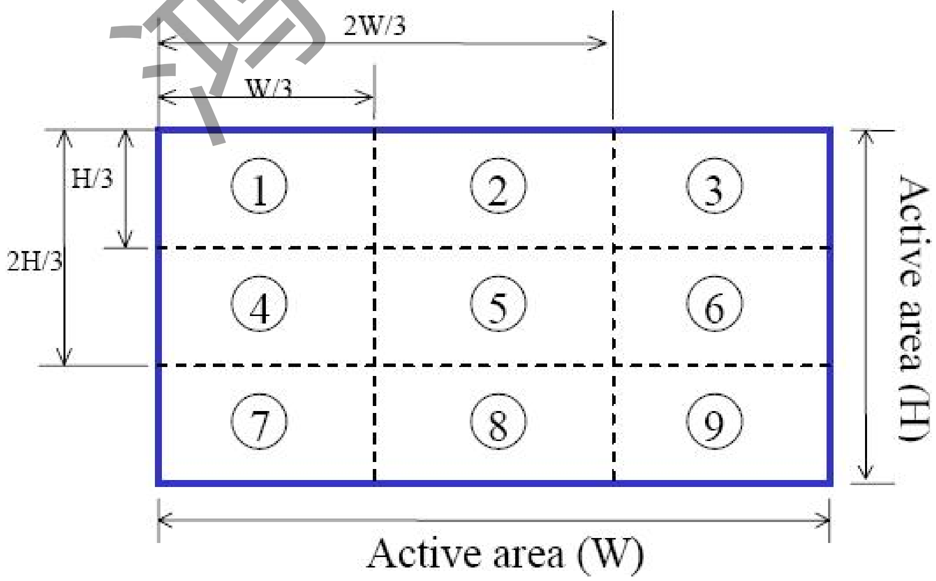
(2) Best Contrast.

★2 Uniform measure condition:

(1) Measure 9 point. Measure location show below;

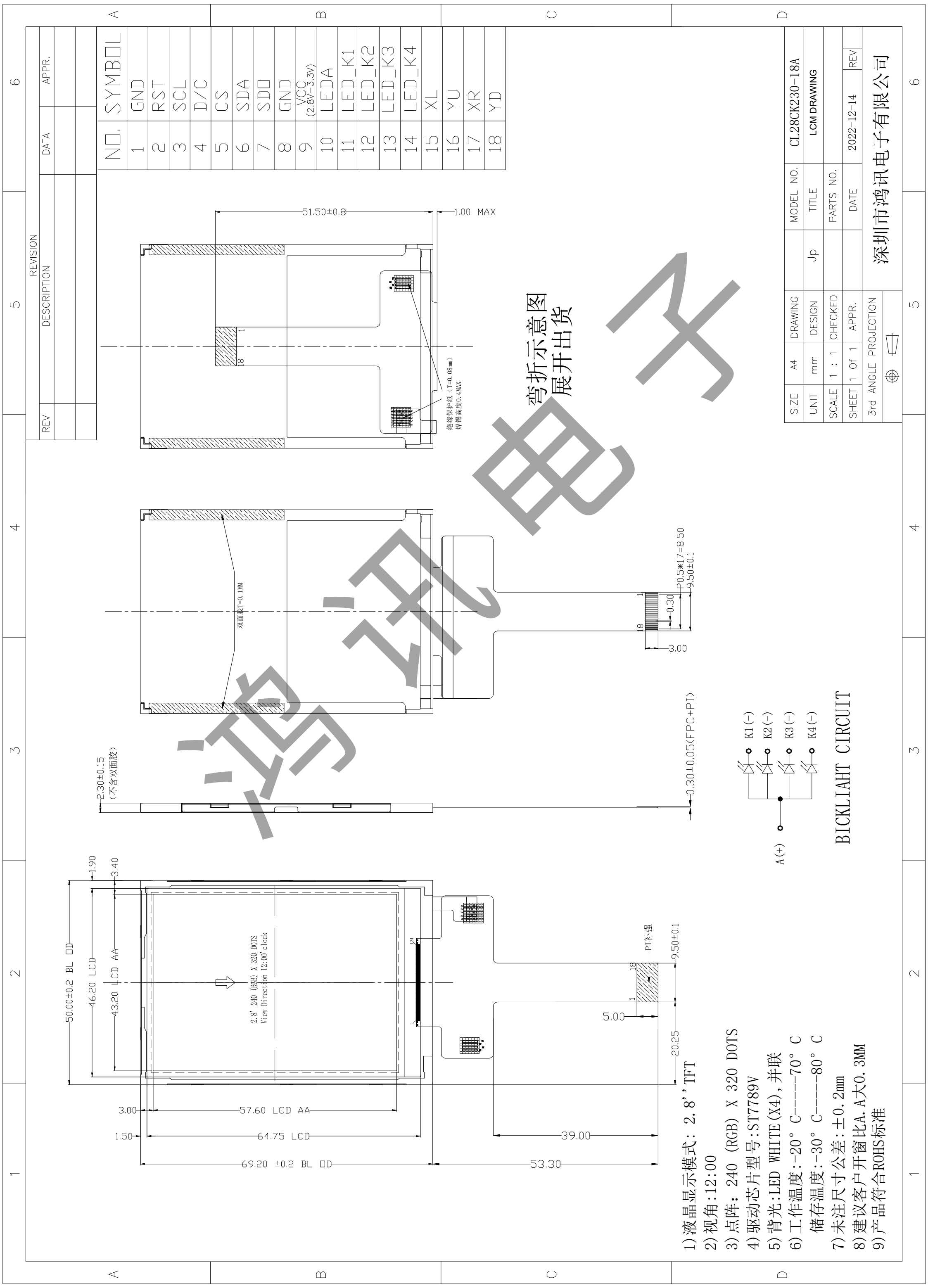
(2) Uniform=(Min. brightness /Max.brightness)*100%

(3)Best Contrast.

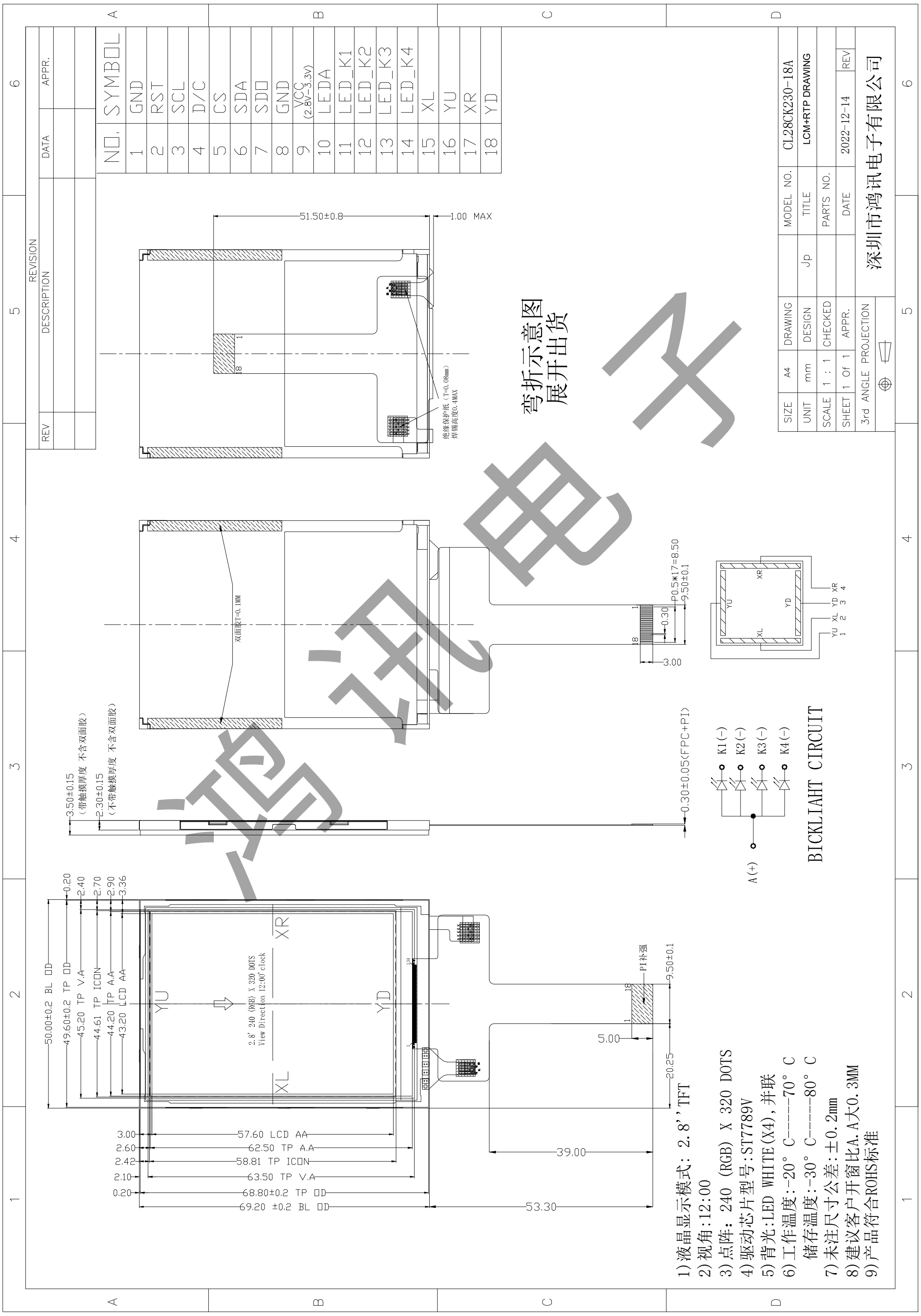


3.0 MECHANICAL OUTLINE DIMENSION

3.1 LCM Drawing without RTP (显示屏不带电阻触摸的图纸)



3.2 LCM Drawing With RTP （显示屏带电阻触摸的图纸）

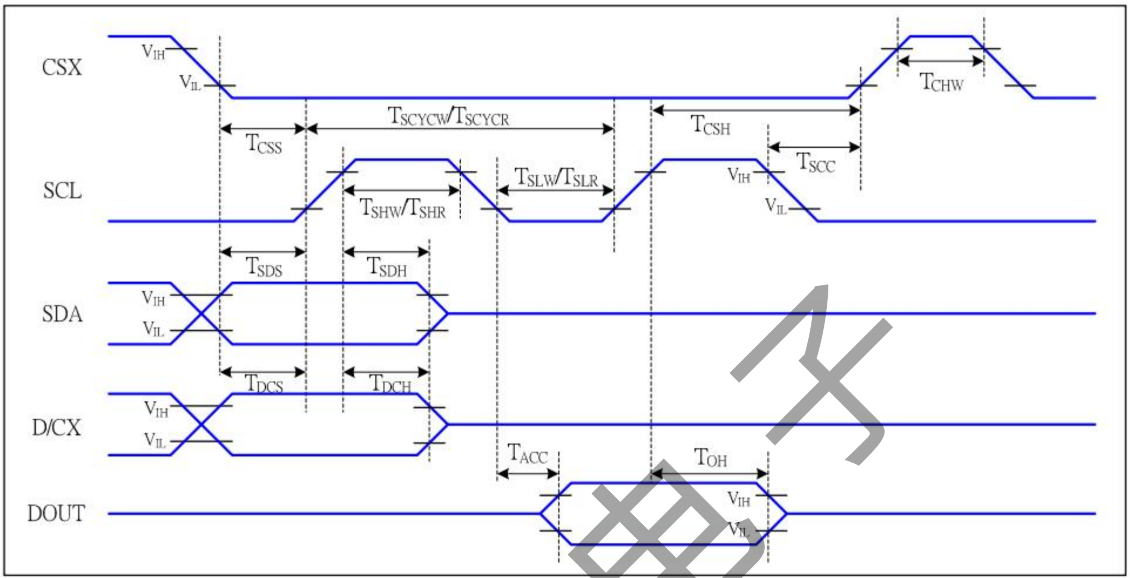


4.0 PIN CONFIGURATION

Pin No.	Symbol	Description
1	GND	Ground (接地脚)
2	RESET	Reset Signal （复位脚）
3	SCL	Serial Interface Clock Pin （串口时钟输入脚）
4	D/C	Display Data/Command Selection Pin （数据或寄存器选择脚）
5	CS	Chip Selection Pin （片选信号）
6	SDA	Serial interface input pin （串口数据输入脚）
7	SDO	Serial interface output pin （串口数据输出脚）
8	GND	Ground (接地脚)
9	VCC	Power Supply(2.8-3.3V) (电源脚)
10	LEDA	Backlight Led Anode Input Pin (+) （背光正极供电脚）
11	LED_K1	Backlight Led Cathode Input Pin (-) （背光负极供电脚）
12	LED_K2	Backlight Led Cathode Input Pin (-) （背光负极供电脚）
13	LED_K3	Backlight Led Cathode Input Pin (-) （背光负极供电脚）
14	LED_K4	Backlight Led Cathode Input Pin (-) （背光负极供电脚）
15	XL	Touch Panel Control Pin. (触摸屏控制脚)
16	YU	Touch Panel Control Pin. (触摸屏控制脚)
17	XR	Touch Panel Control Pin. (触摸屏控制脚)
18	YD	Touch Panel Control Pin. (触摸屏控制脚)

5.0 SIGNAL TIMING SPECIFICATION

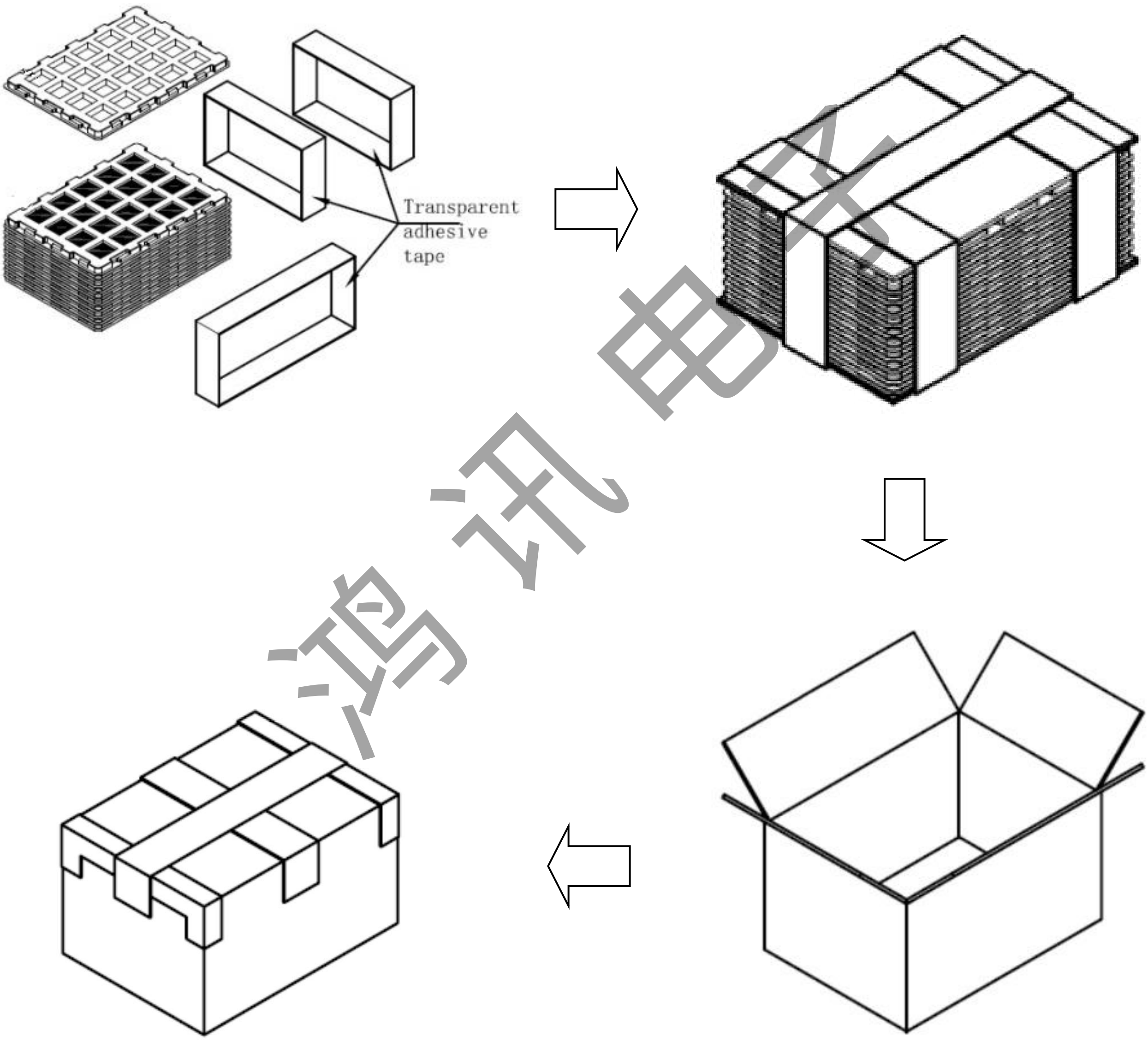
Serial Interface Characteristics (4-line serial)



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15		ns	
	T_{CSH}	Chip select hold time (write)	15		ns	
	T_{CSS}	Chip select setup time (read)	60		ns	
	T_{SCC}	Chip select hold time (read)	65		ns	
	T_{CHW}	Chip select "H" pulse width	40		ns	
SCL	T_{SCYCW}	Serial clock cycle (Write)	16		ns	-write command & data ram
	T_{SHW}	SCL "H" pulse width (Write)	7		ns	
	T_{SLW}	SCL "L" pulse width (Write)	7		ns	
	T_{SCYCR}	Serial clock cycle (Read)	150		ns	-read command & data ram
	T_{SHR}	SCL "H" pulse width (Read)	60		ns	
	T_{SLR}	SCL "L" pulse width (Read)	60		ns	
D/CX	T_{DCS}	D/CX setup time	10		ns	
	T_{DCH}	D/CX hold time	10		ns	
SDA (DIN)	T_{SDS}	Data setup time	7		ns	
	T_{SDH}	Data hold time	7		ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Note: Ta = 25 °C, VDDI=1.65V to 3.3V, VDD=2.5V to 3.3V, AGND=DGND=0V

6.0 PACKING INFORMATION



7.0 RELIABILITY TEST

The Reliability test items and its conditions are shown in below.

No	Test Items	Conditions
1	High temperature storage test	Ta = 80℃, 96 hrs
2	Low temperature storage test	Ta = -30℃, 96 hrs
3	High temperature & high humidity operation test	Ta = 40℃, 90%RH, 48 hrs
4	High temperature operation test	Ta = 70℃, 96 hrs
5	Low temperature operation test	Ta = -20℃, 96 hrs
6	Thermal shock	Ta = -20℃ ↔ 70℃ (2 hr), 30 cycle

8.0 INSPECTION STANDARDS

8.1 AQL(Acceptable Quality Level

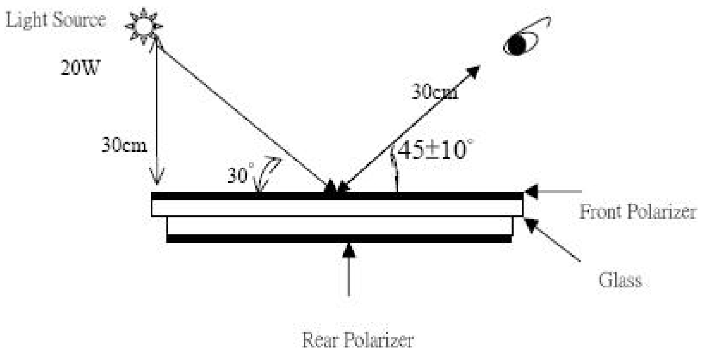
AQL of major and minordefect.

	MAJOR DEFECT	MINOR DEFECT
AQL	0.65	1.5

8.2 Basic conditions forinspection

The LCM face to us, in normal environment, the lux is 1000±200.(Darkroom's lux: 100±50),About anangle of incidence 30,a distance of 30 cm with an angle of 45 degree to check the products without uncovering the film!

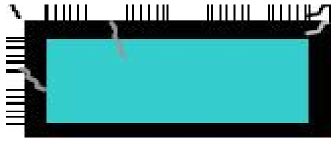
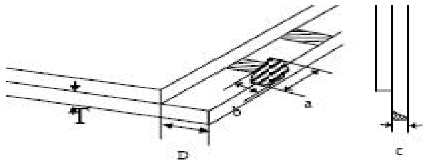
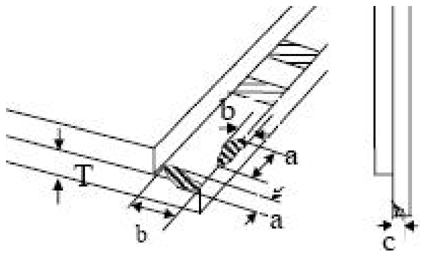
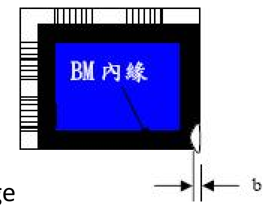
(As shown below)



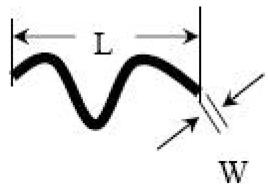
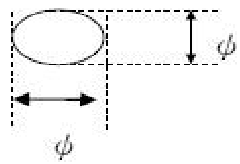
8.3 Inspection item andcriteria

8.3.1 Visual inspection criterion inimmobility

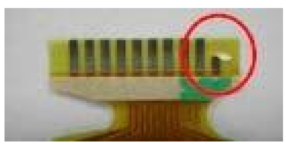
(1) Glassdefect

NO	Defect item	Criteria	Remark
1	Dimension Unconformity (Major defect)	By Engineering Drawing	
2	Cracks (Major defect)	1. Linear crackspanel 【Reject】 2. Nonlinear crack contrastby limited sample	
3	Glass extrude the conductive area (minor defect)	a: disregards and no influence assemblage. 1) b≤1/3Pin width(non bonding area) 【Accept】 2)bonding area≤0.5mm 【Accept】	A: Length, b: Width
4	Pin-side ,conductive area damaged (minor defect)	(a c: disregards) b≤1/3of effective length for bonding electrode 【Accept】	a: length, b: Width, c: Thickness 
5	Pin-side,non-conductive area damaged (minor defect)	1)Damage area don't touch theITO (Inclueling contraposition mark, except scribingmark) 【Accept】 2)C<T b≦BM1/3ofwidth 【Accept】 3)c=T b not touch the seal glue 【Accept】 4)a disregards	a: Length, b: Width c: Thickness 
6	Non-pin-side damage (minor defect)	c<T 1)b exceeds 1/3Bm 【Reject】 c=T b not touch the seal glue 【Reject】	c: Thickness b: width of  damage

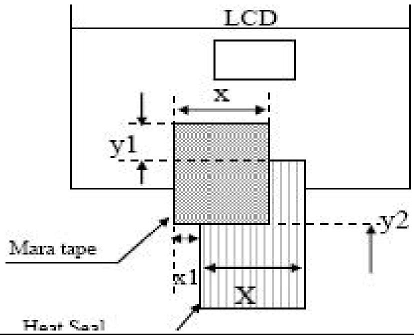
(2) LCD appearance defect(View area)

NO	Defect item	Criteria		Remark
1	Fiber 、 glass cratch、polarizer scratch/folded (minor defect)	Specification	Allowable	note1:L: Length, W: Width note2: disregard if out of AA 
		W≦0.03mm	disregard	
		0.03mm<W≦0.05mm; L≦3.0mm	2	
		0.05mm<W≦0.1mm; L≦3.0mm	1	
		W>0.1mm;L>3.0mm	0	
2	Polarizer bubble 、 concave and convex (minor defect)	ϕ≦0.2mm	disregard	note1: ϕ=(L+W)/2,L:Length, W :Width note2:disregard if out of AA
		0.2mm<ϕ≦0.3mm	2	
		0.3mm<ϕ≦0.5mm	1	
		0.5mm<ϕ	0	
3	Blackdots 、dirtydots 、impurities、eyewinker (minordefect)	ϕ≦0.15mm	disregard	note2:disregard if out of AA 
		0.15mm<ϕ≦0.25mm	2	
		0.25mm<ϕ≦0.3mm	1	
		0.3mm<ϕ	0	
4	Polarizer prick (minor defect)	ϕ≦0.1mm	disregard	note1: ϕ=(L+W)/2,L=Length, W=Width note2:the distance between two dots>5mm
		0.1mm<ϕ≦0.25mm	3	
		ϕ>0.25mm	0	

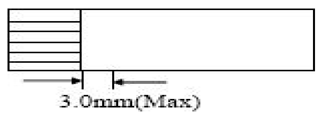
(3) FPC

NO	Defect item	Criteria		Remark
1	Copper screen peel (minor defect)	Copper screen peel 【Reject】		
2	No release tape or peel	No release tape or peel 【Reject】		
3	Dirty dot and impurity of FPC for customer using side (minor defect)	Specification	Allowable	Note1: Cannot have stride ITOimpurities
		ϕ≦0.25mm	2	
		ϕ>0.25	0	

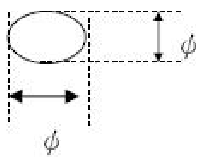
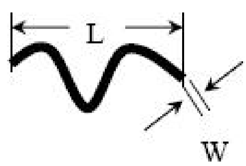
(4) Back tape &Mara tape

NO	Defect item	Criteria	Remark
1	FPC or H/S black tape (minor defect)	1. shiftspec: 1) glue to the polarize 【Reject】 2)ICbare 【Reject】 2. left-and-rightspec: 1) exceed of FPC edge orH-Sedge【Reject】 2) ICbare 【Reject】	
2	No black tape (major defect)	No black tape 【Reject】	
3	Tape position mistake (minor defect)	Not by engineering drawing	
4	Mara tape defect (minor defect)	Peel before pulling the protecting film 【Reject】	

(5) Silicon and Taffy glue

NO	Defect item	Criteria	Remark
1	Quantity of silicon (major defect)	Uncover the ITO and circuit area 【Reject】	note: compared by engineering
2	Taffy glue (major defect)	1.Uncoverthe reveal copperarea【Reject】 2.Cover layer0.3mm(Min)~3.0mm(Max) 【Reject】	note: if customer has special requirement, refer to the technical document 
3	Depth of glue covering (major defect)	Depth of glue covering overtop front Polarizer 【Reject】	Except of the special requirement

8.3.2 Electricalcriteria

NO	Defect item	Criteria		Remark
1	No display (major defect)	No display 【Reject】		
2	Missing line (major defect)	Missing line 【Reject】		
3	Seg-com light and dark (major defect)	Seg-com light and dark 【Reject】		ND filter 2% test
4	No display in immobility (major defect)	No display in immobility 【Reject】		
5	Flicker of Pattern (major defect)	Flicker of Pattern 【Reject】		
6	Mura (major defect)	ND filter 2%test		
7	Over current (major defect)	Over current 【Reject】		
8	Voltage out of specification (major defect)	Voltage out of specification 【Reject】		
9	Pattern blur, error code (major defect)	Pattern blur, error code 【Reject】		
10	Dark light, Flicker (major defect)	Dark light, Flicker 【Reject】		
11	Black/white dots、Dirty dots、eyewinker (major defect)	Specification	Allowable	Note1:disregard if out of AA 
		ϕ≦0.15mm	disregard	
		0.15mm<ϕ≦0.25mm	2	
		0.25mm<ϕ≦0.3mm	1	
		0.3mm<ϕ	0	
12	Fiber glasscrutch Polarizer scratch/folded (major defect)	W≦0.03mm	disregard	Note1:L: Length, W: Width Note2: disregard if out of AA 
		0.03mm<W≦0.0.05mm L≤3.0mm	2	
		0.05mm<W≦0.1mm L≤3.0mm	1	
		W>0.1mm;L>3.0mm	0	

9.0 HANDLING & CAUTIONS

(1) Cautions when taking out the module

• Pick the pouch only, when taking out module from a shipping package.

(2) Cautions for handling the module

• As the electrostatic discharges may break the LCD module, handle the LCD module with care. Peel a protection sheet off from the LCD panel surface as slowly as possible.

• As the LCD panel and back - light element are made from fragile glass material, impulse and pressure to the LCD module should be avoided.

• As the surface of the polarizer is very soft and easily scratched, use a soft dry cloth without chemicals for cleaning.

• Do not pull the interface connector in or out while the LCD module is operating.

• Put the module display side down on a flat horizontal plane.

• Handle connectors and cables with care.

(3) Cautions for the operation

• When the module is operating, do not lose clock, enable signals. If any one of these signals is lost, the LCD panel would be damaged.

• Obey the supply voltage sequence. If wrong sequence is applied, the module would be damaged.

(4) Cautions for the atmosphere

• Dew drop atmosphere should be avoided.

• Do not store and/or operate the LCD module in a high temperature and/or humidity atmosphere. Storage in an electro-conductive polymer packing pouch and under relatively low temperature atmosphere is recommended.

(5) Cautions for the module characteristics

• Do not apply fixed pattern data signal to the LCD module at product aging.

• Applying fixed pattern for a long time may cause image sticking.

(6) Other cautions

• Do not disassemble and/or re-assemble LCD module.

• Do not re-adjust variable resistor or switch etc.

• When returning the module for repair or etc., Please pack the module not to be broken.We recommend to use the original shipping packages.

10.0 REVISION HISTORY

Version	Revise record	Date
V0	Original version	2022-12-30